

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – MCQs

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 3 – MCQs
Weightage:	10% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Answer all questions carefully. Each question carries 2 marks. Total: 100 marks.
- Analyze each question and select the correct answer.
- Apply concepts correctly to solve problem-based MCQs.
- Manage your time efficiently to complete all 50 questions.

Question Type	Focus Area	Questions
MCQs	Analyze and apply Divide and Conquer Algorithms for Merge Sort, Quick Sort, Binary Search, and Matrix Multiplication.	Q1 -Q50

SECTION A – MCQs

MCQ

Link :

CO3 AT3- MCQ
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Language: **Python** C C++ Java

QUESTIONS
Q1 ✓
Quick Sort worst-case pivot condition
2 marks

Q2 ✓
Binary Search comparisons for n=1024
2 marks

Q3 ✓
Strassen's algorithm reduced operation
2 marks

Q4 ✓
Strassen's algorithm efficiency condition
2 marks

Q5 ✓
Merge Sort base case for recursion
2 marks

Q6 ✓
Strassen's algorithm complexity
2 marks

Q7 ✓

Q50 Reason Merge Sort is not in-place

2 marks

Merge Sort requires extra memory to store temporary arrays during merging. Why is it not in-place?

A. Uses recursion

B. Needs extra memory

C. Uses comparisons

D. Uses pivot

 **Save Answer**
Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Online Quiz / MCQ Test					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

Mark Evaluations:

No. of MCQ Questions	Correct Answers	Wrong/Not Answers	Total Marks (100)
50			

Faculty In-charge	Course Coordinator	Head of Department
Name: _____ Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____